

Evaluating Depth Estimation on Reflective Surfaces

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1 Introduction

Reconstructing 3D scenes from consumer-grade cameras is receiving increasing attention in computer vision. While low-cost depth-sensing cameras exist, such as the Kinect [1] and Intel RealSense [2] camera families, advances in deep convolutional neural networks have made it possible to estimate depth values directly from RGB images, potentially removing the need for dedicated depth hardware. Monocular depth estimation seeks to infer scene geometry from a single image, enabling applications in robotics, AR, and industrial automation.

Problem Statement Standard self-supervised monocular depth estimation (SS-MDE) models rely on a photometric consistency assumption (view synthesis): a target view can be reconstructed from a source view using predicted depth and camera motion. This implicitly assumes that surface appearance is approximately Lambertian and remains consistent under viewpoint changes [3]. Indoor environments such as factories often contain mirrors, shiny metals, and glossy surfaces that exhibit specular reflection. In these regions, photometric consistency is violated and SS-MDE frequently produces incorrect depth predictions. Our goal is to determine whether predicting per-pixel uncertainty can reduce these failure modes by down-weighting unreliable photometric supervision [4].

Related Work SS-MDE has become popular due to removing the need for dense depth labels; Monodepth2 [3] is a widely used baseline that improves sharpness and robustness via minimum reprojection and multi-scale supervision. However, reflective surfaces remain challenging. Recent directions include:

- **3D distillation:** uses multi-view 3D mesh reconstruction to create improved pseudo-labels and distill them into a student depth network, improving reflective-region accuracy at high computational cost (multiple models + 3D reconstruction) [5].
- **Intrinsic image decomposition:** jointly estimates intrinsic properties (reflectance/shading) to identify non-Lambertian regions and suppress corrupted gradients during depth learning; avoids explicit reflective annotations and expensive ensembles, but still struggles on mirrors and transparent surfaces [6].
- **Segmentation and inpainting:** uses segmentation masks for transparent/mirror (ToM) surfaces and inpaints those regions to generate virtual depth labels for fine-tuning; performance depends heavily on segmentation quality [7].

Uncertainty modeling has also been studied systematically for SS-MDE. Poggi et al. [4] show that uncertainty is measurable in self-supervised depth and can be exploited both to improve predictions (by down-weighting unreliable photometric supervision) and to identify failure cases. More broadly, Kendall and Gal [8] distinguish *aleatoric* uncertainty (data-dependent noise such as specularities, motion blur, or occlusions) from *epistemic* uncertainty (model uncertainty due to limited training coverage), and show that explicitly modeling uncertainty can improve robustness while providing calibrated confidence for downstream decision making. In reflective indoor scenes, many violations of the photometric consistency assumption are inherently aleatoric (e.g., view-dependent highlights on metal and mirrors), making per-pixel uncertainty particularly promising as a mechanism to automatically reduce the influence

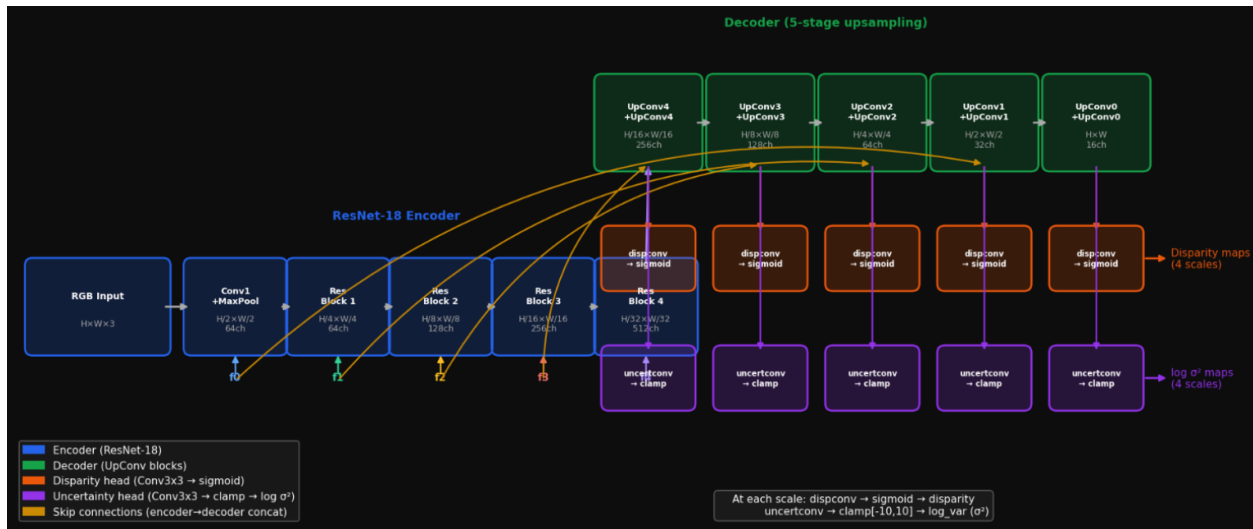


Figure 1: Model Architecture with uncertainty head added

of these corrupted regions during training and to flag unreliable depth at inference time. Related Bayesian deep learning work further supports the practical utility of uncertainty estimates, including Monte Carlo dropout as an approximate Bayesian inference method [9] and earlier demonstrations that deep networks can output heteroscedastic predictive variance [8].

2 Methods

2.1 Datasets and Preprocessing

We use the Booster dataset [10], which contains indoor scenes with many reflective, transparent and specular surfaces. It also provides high resolution stereo image pairs, camera calibration, and dense metric depth, which fits our purpose.

Splits Booster contains 228 images from 38 scenes with lighting variations within scenes. We split *by scene* to reduce leakage across train/val/test. In our experiments:

- Monodepth2 fine-tuning: We split the dataset with a train/val/test ratio of 72:8:20 images

across all 38 scenes with scene-averaged reporting.

- Uncertainty head: training on 182 images from 37 scenes and evaluation on 46 images from 28 scenes.

Exact counts can vary slightly depending on scene-level partitioning and filtering.

Downsampling Original Booster resolution is 4112×3008 . Due to computational limits, we downsampled while preserving aspect ratio using Lanczos resampling. Most runs used 512×384 ; we additionally tested 768×576 and 640×192 to study resolution sensitivity. Camera intrinsics were scaled accordingly, which is standard when resizing calibrated imagery.

2.2 Baseline Model: Monodepth2

We use Monodepth2 [3] with a ResNet-18 encoder and U-Net style decoder. As in Monodepth2, depth is predicted at multiple scales and upsampled for loss computation. We initialize from KITTI-pretrained weights using transfer learning, then fine-tune on

Booster. In stereo-only mode, the pose transform is fixed by the known stereo baseline, reducing degrees of freedom and stabilizing training.

2.3 Uncertainty-Aware Extension

Following Poggi et al. [4], we add a parallel decoder branch to predict per-pixel *log-variance* ($\log \sigma^2$) representing photometric uncertainty. The intuition is that specular or reflective pixels violate the photometric model, so the network should learn to assign them higher uncertainty, thus down-weighting their contribution to the photometric loss.

Architecture We augment the Monodepth2 decoder with an additional convolutional “uncertconv” head that outputs a log-variance map at multiple scales (Figure 1).

Monodepth fine-tuning Loss For standard Monodepth2 fine-tuning, we minimize a photometric reprojection loss plus an edge-aware smoothness regularizer [3]. The photometric error combines SSIM and L1:

$$pe(I_a, I_b) = \frac{\alpha}{2} (1 - \text{SSIM}(I_a, I_b)) + (1 - \alpha) \|I_a - I_b\|_1, \quad (1)$$

where $\alpha = 0.85$ (as used in our implementation and in Monodepth2). The total photometric reprojection loss is

$$L_p = \sum_{t'} pe(I_t, I_{t' \rightarrow t}), \quad (2)$$

where the synthesized target view is obtained by warping the source image using predicted depth and known stereo geometry:

$$I_{t' \rightarrow t} = I_{t'} \langle \text{proj}(D_t, T_{t \rightarrow t'}, K) \rangle. \quad (3)$$

We additionally use the edge-aware disparity smoothness loss:

$$L_s = |\partial_x d_t^*| e^{-|\partial_x I_t|} + |\partial_y d_t^*| e^{-|\partial_y I_t|}, \quad (4)$$

and the overall Monodepth2 fine-tuning objective is

$$L_{\text{mono}} = \mu L_p + \lambda L_s, \quad (5)$$

with μ and λ set following common Monodepth2 practice and tuned on the validation set.

Uncertainty-Modulated Negative Log Likelihood (NLL) Loss To handle reflective and specular regions where photometric consistency is unreliable, we add a second decoder head that predicts per-pixel log-variance $\log \sigma^2$ (photometric uncertainty) [4]. We replace the plain photometric term with an NLL-style objective:

$$L_{\log}(x) = \frac{1}{2} pe(x) e^{-\log \sigma^2(x)} + \frac{1}{2} \log \sigma^2(x), \quad (6)$$

computed per pixel x . The first term down-weights pixels with high predicted uncertainty (e.g., mirrors, specular highlights, occlusions), and the $\frac{1}{2} \log \sigma^2$ term regularizes against the trivial solution of predicting infinite uncertainty everywhere [4, 8]. The final uncertainty-aware objective becomes

$$L_{\text{uncert}} = \text{mean}_x(L_{\log}(x)) + \lambda L_s, \quad (7)$$

where we keep the same edge-aware smoothness term L_s as in Monodepth2 and tune λ on the validation set.

2.4 Training Details

Monodepth2 fine-tuning We fine-tune a KITTI-pretrained Monodepth2 model on Booster using a learning rate of 1×10^{-5} and batch size 8. To stabilize optimization on the small dataset, we freeze the ResNet-18 encoder and update only the decoder weights. We also disable auto-masking during fine-tuning.

Uncertainty-aware training For uncertainty learning, we train in stereo-only mode using the known, fixed stereo baseline transform (i.e., no PoseNet). The uncertainty model is initialized from our Booster fine-tuned Monodepth2 checkpoint and then trained for up to 20 epochs. We select the best model using validation performance; for the 512×384 NLL log-variance configuration, the best checkpoint occurred early (epoch 4). We evaluated two learning rates, 10^{-4} and 2×10^{-5} , and found 2×10^{-5} produced better validation results overall.

Model	RMSE (m)	Abs Rel	$\delta < 1.25$ (a1)
KITTI-pretrained Monodepth2	0.99	0.44	0.49
Fine-tuned on Booster	0.317	0.18	0.67

Table 1: Monodepth2 fine-tuning results (presentation). a2/a3 also improved to 0.90/0.98.

Model	AbsRel	SqRel	RMSE	RMSE _{log}	a1	a2	a3
Baseline (KITTI, no FT)	0.579	0.904	0.830	0.631	0.254	0.487	0.678
FT 512×384 (10 ep)	0.274	0.185	0.449	0.400	0.419	0.711	0.879
FT 768×576 (20 ep)	0.328	0.226	0.507	0.489	0.284	0.608	0.807
NLL log-var 640×192 (ep15)	0.255	0.152	0.402	0.350	0.499	0.810	0.894
NLL log-var 512×384 (ep4)	0.244	0.151	0.390	0.331	0.533	0.800	0.911

Table 2: Depth results with uncertainty enabled (presentation).

2.5 Evaluation Metrics

We report depth estimation performance using the standard error and accuracy metrics commonly adopted in monocular depth estimation [3]. Let d denote the predicted depth and d^* the ground-truth depth over N valid pixels. We measure error using Abs Rel, Sq Rel, RMSE, and RMSE_{log} (lower is better). In addition, we report threshold accuracies $\delta < 1.25$, $\delta < 1.25^2$, and $\delta < 1.25^3$ (higher is better), where a prediction is considered correct if

$$\max\left(\frac{d}{d^*}, \frac{d^*}{d}\right) < \delta. \quad (8)$$

Uncertainty quality To evaluate whether the predicted uncertainty is informative of depth errors, we follow the sparsification protocol of Poggi et al. [4]. Specifically, pixels are ranked by predicted uncertainty and progressively removed from the evaluation set; the depth error (e.g., Abs Rel) is recomputed on the remaining pixels. A well-calibrated uncertainty estimate should yield a faster decrease in error compared to random removal and approach the oracle ordering (removing the highest-error pixels first). We summarize this behavior using AUSE (area under the sparsification error curve, lower is better) and AURG (area under the random gain curve, higher is better) [4].

3 Results

3.1 Monodepth2 Fine-tuning on Booster

We first evaluate the effect of domain adaptation by fine-tuning a KITTI-pretrained Monodepth2 model on the Booster dataset. Quantitative results on our held-out Booster test split (scene-averaged over 54 images across 8 scenes) are reported in Table 1. Fine-tuning yields a substantial improvement over the KITTI-pretrained baseline, reducing RMSE from 0.99m to 0.317m and Abs Rel from 0.44 to 0.18, while increasing $\delta < 1.25$ accuracy from 0.49 to 0.67.

Qualitatively, the fine-tuned model better captures indoor scene layout and object boundaries compared to the pretrained baseline, reflecting improved adaptation to Booster’s reflective indoor domain. Figure 2 illustrates a representative example where object edges are sharper after fine-tuning, although noticeable high-frequency artifacts remain, particularly around reflective or low-texture regions. Overall, these results indicate that fine-tuning on in-domain data is critical for SS-MDE in reflective indoor environments, but non-Lambertian surfaces continue to pose challenges.

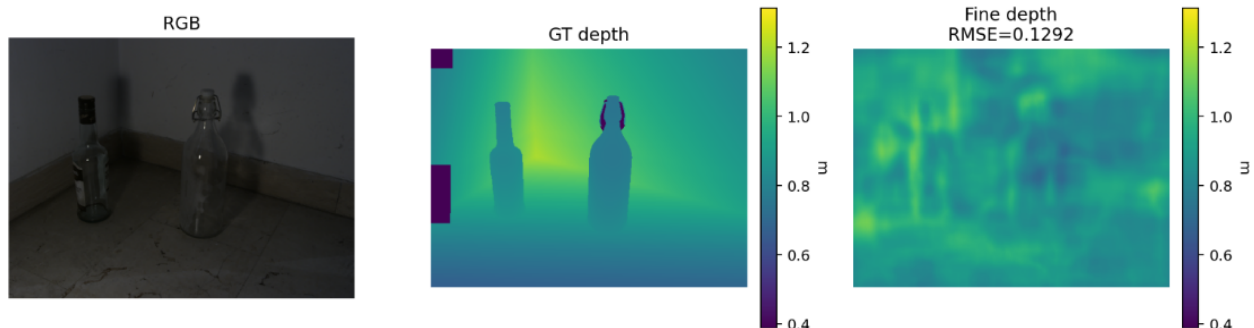


Figure 2: Fine-tuned example (bottle scene). Left: RGB input; Middle: ground-truth depth; Right: predicted depth.

3.2 Uncertainty-Enabled Results

We next assess whether explicitly predicting per-pixel photometric uncertainty improves robustness on reflective indoor scenes. We experimented with several input resolutions and uncertainty-training configurations. The best performing uncertainty-aware model uses the log-variance NLL formulation at 512×384 , and is selected via validation performance (best checkpoint at epoch 4). We report its test performance over 46 images from 28 scenes in Table 2.

Relative to a Booster fine-tuned baseline at comparable resolution, introducing uncertainty improves Abs Rel from 0.274 to 0.244 (approximately 11% relative) and increases $\delta < 1.25$ accuracy (a1) from 0.419 to 0.533. These gains suggest that uncertainty modulation can partially mitigate violations of the photometric consistency assumption by reducing the contribution of unreliable pixels during optimization.

Qualitatively, the predicted uncertainty maps frequently highlight non-Lambertian regions such as mirrors, chrome fixtures, and glass, despite the absence of explicit reflective-surface annotations. As illustrated in Figure 3, high uncertainty is concentrated on reflective areas, which effectively down-weights their influence in the uncertainty-modulated photometric loss and can lead to more stable depth predictions in challenging regions.

3.3 Uncertainty Calibration via Sparsification

To assess whether the predicted uncertainty is a reliable proxy for depth error, we follow the sparsification-based evaluation protocol in [4]. Table 3 reports AUSE (lower is better) and AURG (higher is better) for our best log-variance NLL model and an MC-dropout baseline.

Method	AUSE ↓	AURG ↑
NLL log-var (best)	0.102	−0.019
MC Dropout	0.146	−0.024

Table 3: Uncertainty sparsification metrics.

While uncertainty modeling improves overall depth accuracy (Table 2), the negative AURG values indicate that the uncertainty ranking is not consistently better than random at identifying high-error pixels. This suggests that the learned uncertainty is capturing photometric unreliability (e.g., specularities or view-dependent effects) that does not always translate monotonically into depth error, particularly in reflective indoor scenes.

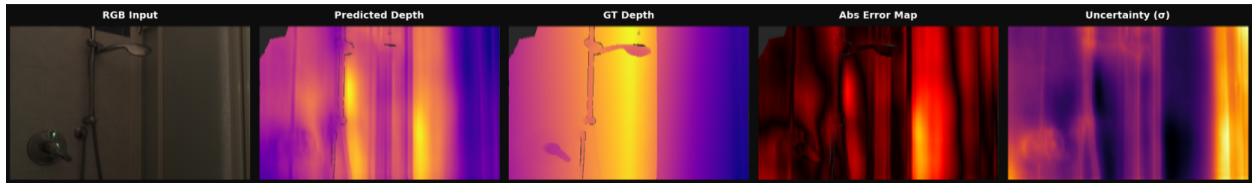


Figure 3: Uncertainty qualitative example (Shower scene). From left to right: RGB input, predicted depth, ground-truth depth, absolute error map and predicted uncertainty.

4 Discussion

The experimental results highlight the sensitivity of self-supervised monocular depth estimation to non-Lambertian appearance. In reflective indoor scenes, view-dependent effects violate the photometric consistency assumptions underpinning view synthesis training, leading to systematic depth artifacts [3]. Domain adaptation via fine-tuning on Booster substantially improves performance relative to KITTI pretraining, indicating that in-domain supervision signals and scene statistics are important for indoor reflective environments. Moreover, augmenting Monodepth2 with a log-variance uncertainty head yields further gains in both error and accuracy metrics, supporting the hypothesis that uncertainty-modulated objectives can reduce the impact of unreliable photometric supervision [4, 8]. Beyond accuracy improvements, the predicted uncertainty maps provide an interpretable indication of where the model is likely to be unreliable, which is desirable in downstream applications that require risk-aware decision making.

Limitations Several limitations remain. First, while uncertainty improves aggregate metrics, qualitative inspection suggests that fine object boundaries can become less sharp and may exhibit increased high-frequency artifacts; additional regularization or edge-preserving refinement could improve detail reconstruction. Second, we only evaluated a limited set of uncertainty mechanisms (log-variance NLL and MC dropout); other approaches such as ensembles, snapshot-based approximations, or alternative aleatoric formulations may provide better calibration and ranking of errors [4, 9]. Third, sparsifi-

cation results indicate that photometric uncertainty does not consistently correlate with depth error in reflective regions, implying that the learned uncertainty may reflect appearance unreliability rather than geometric error. Finally, higher input resolution did not monotonically improve performance in our setting, suggesting interactions between resolution, warping accuracy, batch size constraints, and optimization stability that warrant further investigation.

Identifying failure Modes on Reflective Surfaces A complete failure-mode study with region-specific quantitative evaluation is left to future work. Nevertheless, based on qualitative inspection of Booster scenes and known limitations of photometric self-supervision under non-Lambertian appearance, we highlight three common failure modes that motivate uncertainty-aware training.

- **Mirror surfaces:** Mirror reflections can be interpreted as real geometry behind the mirror, leading the model to predict depth “inside” the reflective plane. Since the reflected content can appear visually consistent across viewpoints, a photometric reconstruction objective may still be minimized despite incorrect 3D structure, producing large local depth errors and artificial discontinuities.
- **Transparent/clear surfaces:** For glass or other transparent objects, SS-MDE may predict the depth of the background observed through the surface rather than the surface itself, because the photometric signal is dominated by what is visible behind the object. Additionally, reflections on transparent materials introduce view-

dependent appearance changes that violate the brightness constancy assumption, further degrading supervision.

- **Specular highlights:** Specular highlights on metallic or glossy surfaces are strongly view-dependent and can shift across the surface as the camera moves. These highlight motions violate photometric consistency and can generate spurious depth spikes, oversmoothing, or texture-copy artifacts near high-intensity regions.

Future Work and Applications Uncertainty estimates are useful not only for improving depth optimization but also for enabling risk-aware perception in downstream systems. Future work includes (i) more detailed failure-mode evaluation with reflective-region masks, (ii) exploring alternative uncertainty formulations and calibration methods (e.g., ensemble- or snapshot-based epistemic uncertainty), and (iii) integrating uncertainty into decision policies. Potential applications include robotic manipulation in industrial environments, where high-uncertainty depth regions can trigger slower or more conservative toolpaths when shiny parts, glare, or motion blur reduce reliability; assistive navigation for visually impaired users, where uncertainty can suppress misleading depth cues from glass or mirrors and prompt re-aiming for additional views; and medical or assistive robotics in specular settings, where uncertainty can encourage conservative motion unless depth confidence is sufficient.

5 Conclusion

We developed an uncertainty-aware self-supervised monocular depth estimation pipeline by augmenting Monodepth2 with a log-variance NLL uncertainty head and fine-tuning on the Booster dataset, which contains indoor scenes with frequent reflective surfaces. Fine-tuning on Booster substantially improved performance over KITTI pretraining, and incorporating uncertainty further improved depth accuracy, achieving a best Abs Rel of 0.244 and $\delta < 1.25$ (a1) of 0.533. In addition, the uncertainty head quali-

tatively highlights reflective regions without requiring explicit reflective-surface annotations. Nonetheless, sparsification-based analysis indicates that the learned photometric uncertainty does not always rank depth errors better than random, suggesting that improved calibration and stronger coupling between uncertainty and geometric error are important directions for future work in non-Lambertian environments.

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